

MEPCO SCHLENK ENGINEERING COLLEGE (Autonomous), SIVAKASI
Department of Information Technology
Assessment Test
Integrated Technical Training

Class : VI Sem (Common to EEE, ECE and BME) Max. Marks : 50
Date : 20-06-2026 Time : 3.15 PM to 4:45 PM

PART-A [5 X 10 =50 marks]

1. Write a Java program that accepts a number containing at least three digits and repeatedly finds the sum of its digits until a single-digit number is obtained.

Sample Input

987

Process

$9 + 8 + 7 = 24$

$2 + 4 = 6$

Sample Output

Single Digit: 6

2. Write a Java program to toggle the case of each character in a given string. Convert uppercase letters to lowercase and lowercase letters to uppercase.

Sample Input

HeLLo WoRLd

Sample Output

hElLO wOrlD

3. Write a Java program to create a class named Calculator with the following static methods:

- add(int a, int b)
- subtract(int a, int b)
- multiply(int a, int b)
- divide(int a, int b)

Write a main() method to call these methods without creating an object of the Calculator class.

Sample Input

a = 20

b = 5

Sample Output

Addition: 25

Subtraction: 15

Multiplication: 100

Division: 4

4. Write a Java program to develop a cab booking application where customers can book rides in different ways using method overloading:

1. By pickup location and destination.
2. By pickup location, destination, and vehicle type.
3. By pickup location, destination, vehicle type, and coupon code.

Display the booking details based on the selected option.

Sample Input

Booking 1:

Pickup: Madurai

Destination: Sivakasi

Booking 2:

Pickup: Madurai

Destination: Chennai

Vehicle Type: Sedan

Booking 3:

Pickup: Madurai

Destination: Trichy

Vehicle Type: SUV

Coupon Code: SAVE20

Sample Output

Booking Confirmed

Pickup: Madurai

Destination: Sivakasi

Booking Confirmed

Pickup: Madurai

Destination: Chennai

Vehicle Type: Sedan

Booking Confirmed

Pickup: Madurai

Destination: Trichy

Vehicle Type: SUV

Coupon Code Applied: SAVE20

5. Develop a Java program to track the capacity of a parking lot with the following requirements:

- Total parking slots = 100
- Each car entering the parking lot reduces the number of available slots.
- The available slots should be shared globally among all parking lot objects using a static variable.

Display the number of available parking slots after each car enters.

Sample Input

Cars Entering: 3

Sample Output

Car 1 Entered

Available Slots: 99

Car 2 Entered

Available Slots: 98

Car 3 Entered

Available Slots: 97

1.